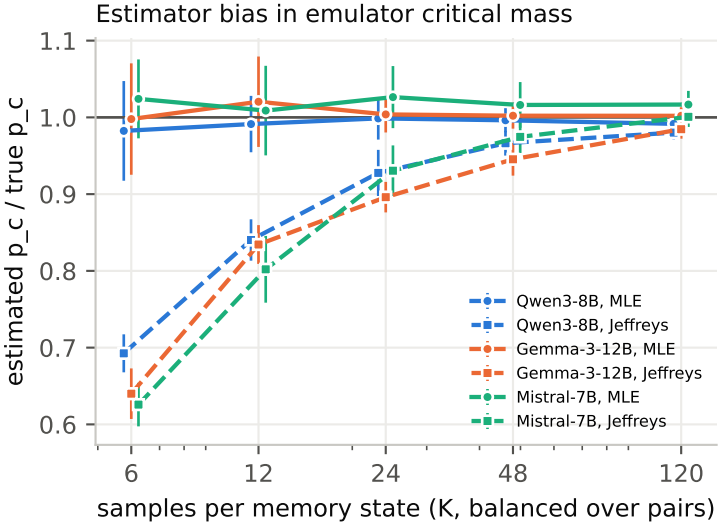
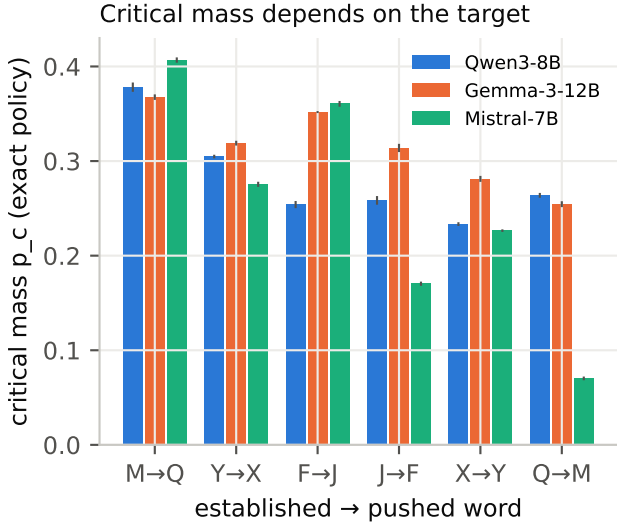

Estimator bias in emulator critical mass
estimated p_c / true p_c
1.1
1.0
0.9
0.8
0.7
0.6
6
12
24
48
120
samples per memory state (K, balanced over pairs)
Qwen3-8B, MLE
Qwen3-8B, Jeffreys
Gemma-3-12B, MLE
Gemma-3-12B, Jeffreys
Mistral-7B, MLE
Mistral-7B, Jeffreys


Critical mass depends on the target
critical mass p_c (exact policy)
0.4
0.3
0.2
0.1
0.0
Qwen3-8B
Gemma-3-12B
Mistral-7B
M→Q
Y→X
F→J
J→F
X→Y
Q→M
established → pushed word